L'Hôpital's Rule (variant)

If $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = \infty$ & $\lim_{x\to\infty} \frac{f'(x)}{g'(x)} = \ell$

then $\lim_{x\to\infty} \frac{f(x)}{g(x)} = \ell$

Pf sketch Apply CMVT to f, g on (N, x), N is very large $x \to \infty$

$\exists\, y \in (N, x)$ s.t. $f'(y)(g(x) - g(N)) = g'(y)(f(x) - f(N))$

For N large enough, we can rewrite

$$\frac{f'(y)}{g'(y)} = \frac{f(x) - f(N)}{g(x) - g(N)}$$

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(N)}{g(x) - g(N)} \cdot \frac{f(x)}{f(x) - f(N)} \cdot \frac{g(x) - g(N)}{g(x)}$$

From assumption,

$$\lim_{x\to\infty} \frac{f(x)}{f(x) - f(N)} = \lim_{x\to\infty} \frac{g(x) - g(N)}{g(x)} = 1$$

$f(x) - f(N) \neq 0$ for x large, b/c $\lim_{x\to\infty} f(x) = \infty$, $f(N)$ is a fixed #

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{f(x) - f(N)}{g(x) - g(N)} \cdot \lim_{x\to\infty} \frac{f(x)}{f(x) - f(N)} \cdot \lim_{x\to\infty} \frac{g(x) - g(N)}{g(x)}$$

$$= \lim_{x\to\infty} \frac{f'(y)}{g'(y)} \cdot 1$$

for some $y \in (N, x)$

$$= \lim_{x\to\infty} \frac{f'(x)}{g'(x)}, \quad y \in (N, x) \text{ so } y \to \infty \text{ if } N \to \infty$$

$$= \ell$$

∎

XII. Inverse Functions

Recall $f: \mathbb{R} \to \mathbb{R}$ ($D \to \mathbb{R}$) is a function, written as $f = \{(x,y); y = f(x)\}$

If $(a,b), (a,c) \in f$, then $b = c$

Def A function f is injective if $f(a) \neq f(b)$ whenever $a \neq b$.

Ex ① $f(x) = x$ is injective

Pf Suppose that $f(a) = f(b)$

Since $f(a) = a$, $f(b) = b$

$\Rightarrow a = b$

② $f(x) = C$ is not injective

Want $a \neq b$ s.t. $f(a) = f(b)$

③ $f(x) = x^2$ is not injective

$f(1) = f(-1)$ although $1 \neq -1$

④ $f(x) = x^2$, $D = [0, \infty)$ is injective

Pf Suppose $f(a) = f(b)$ WTS $a = b$

By def, $a^2 = b^2$

Using $a, b \geq 0$, we conclude $a = b$.

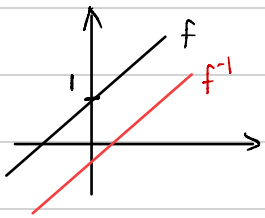
Def If $f = \{(x,y) : f(x) = y\}$ is a function, its inverse $f^{-1} = \{(y,x) : f(y) = x\}$

Note f^{-1}, by this definition, is not necessarily a function.

Ex ① If $f(x) = x + 1$

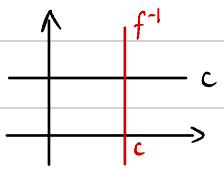
$f = \{(x, x+1) : x \in \mathbb{R}\}$

$f^{-1} = \{(x+1, x) : x \in \mathbb{R}\}$



② $f(x) = c$ $\quad f(x) = \{(x, c) : x \in \mathbb{R}\}$

$f^{-1} = \{(c, x) : x \in \mathbb{R}\}$



Prop f^{-1} is a function iff f is injective

Pf ($\Rightarrow$) Suppose f^{-1} is a function

(If $(a,b), (a,c) \in f^{-1}$, then $b = c$)

WTS: f is injective

(If $(b,a), (c,a) \in f$, then $b = c$)

If $(b,a), (c,a) \in f$, this means $(a,b), (a,c) \in f^{-1}$

Since f^{-1} is a function, then $b = c$

So f is injective

($\Leftarrow$) Assume f is injective

By def, whenever $(b,a), (c,a) \in f \Rightarrow b = c$

That means $(a,b), (a,c) \in f^{-1}$

Therefore, f^{-1} is a function.

∎

Properties of f^{-1}

① $(f^{-1})^{-1} = f$

Idea If $(a,b) \in f$, $(b,a) \in f^{-1}$, $(a,b) \in (f^{-1})^{-1}$

To prove: ① $f \subseteq (f^{-1})^{-1}$

② $(f^{-1})^{-1} \subseteq f$

② If $f(a) = b$, then $f^{-1}(b) = a$
$(a,b) \in f$ $\quad$ $(b,a) \in f^{-1}$

③ If f, f^{-1} are functions, then $f \circ f^{-1} = f^{-1} \circ f = I = \{(a,a) : a \in \mathbb{R}\}$

$f: D \to \mathbb{R}$ $\quad$ $f^{-1}: \mathbb{R} \to D$

Pf ① $f \circ f^{-1} \subseteq I_R$

② $I_R \subseteq f \circ f^{-1}$

① If $(x,y) \in f \circ f^{-1} \iff y = f \circ f^{-1}(x) = f(f^{-1}(x)) = x$ (by ②)
(WTS $y = x$)

② $I_R \subseteq f \circ f^{-1}$

Thm If f is cts. & injective on an interval I, then f is strictly monotone.

Pf. Let $a_0 \neq b_0 \in I$. Since f is injective $\Rightarrow f(a_0) \neq f(b_0)$

WoLoG assume $f(a_0) < f(b_0)$

WTS: f is strictly increasing on all of I

Idea Let $x_t = (1-t)a_0 + ta_1$

$y(t) = (1-t)b_0 + tb_1$

As time progresses from 0 to 1, x_t traces out a_0 to a_1

y_t traces out b_0 to b_1

$x_t < y_t \quad \forall t \in [0,1]$

Define $g(t) = f(y_t) - f(x_t)$

WTS $g(t) > 0$

x_t, y_t, f are all cts.

$\Rightarrow g(t)$ cts.

$g(0) = f(y_0) - f(x_0) = f(b_0) - f(a_0) > 0$

$g(t) \neq 0$ b/c f is injective & $x_t \neq y_t$

By IVT, $g(t) > 0$

$\Rightarrow f$ is increasing b/c a_0, a_1, b_0, b_1 are arbitrary.

